

# Data-driven Remaining Useful Life Prediction for Optical Modules in Intelligent Computing Networks

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**Abstract**—The stability of optical modules significantly impacts overall system performance in intelligent computing networks. Failures in optical modules can lead to a series of issues, diminishing the operational efficiency of intelligent computing centers. Therefore, accurately predicting the remaining useful life of optical modules is crucial for maintaining network stability and minimizing maintenance costs. Currently, traditional physics-based prediction methods rely heavily on precise parameters and are therefore unsuitable for complex intelligent computing network environments. In contrast, existing data-driven prediction models typically use simulated or small-scale experimental data, resulting in limited model generalization capabilities. This paper collects and analyzes the full-lifecycle data of all optical modules in a real intelligent computing network environment, and uses LSTM and Transformer to construct prediction models for forecasting the remaining useful life of optical modules. This provides a technical solution for the intelligent operation and maintenance of optical modules in intelligent computing networks.

**Keywords**—Intelligent computing networks, optical module, remaining useful life, fault prediction.

## I. INTRODUCTION

With the rapid advancement of artificial intelligence technologies, intelligent computing centers have emerged as critical infrastructure driving innovation and transformation across various industries [1]. Within these centers, since optical modules are the core components responsible for high-speed, high-capacity data transmission, which are crucial for ensuring the overall performance and reliability of the system [2]. Large-scale intelligent computing centers typically deploy tens of thousands of optical modules. A single module failure can cause a cascade of problems, including data transmission interruptions, computational delays, or task failures, significantly impacting operational efficiency and service quality. Operational statistics show that in AI clusters with tens of thousands of accelerator cards, maintenance staff address an average of over three optical module-related issues daily, with

each issue taking about 2 to 4 hours to resolve, primarily due to fault localization.

Over time, the performance of optical modules gradually degrades, evolving from a “sub-healthy” state to eventual failure. Although this degradation process may not immediately trigger alarms, it can cause fluctuations in network performance that adversely affect user experience. Consequently, accurately predicting the lifespan of optical modules and implementing preemptive maintenance strategies are crucial for ensuring the continuous and stable operation of intelligent computing center networks, reducing maintenance costs, and enhancing the overall quality of AI computing services.

Currently, extensive research has been conducted in the field of optical module lifetime prediction. Some scholars have adopted traditional physics-based approaches, constructing predictive models by analyzing changes in the physical characteristics of optoelectronic components within the module, such as laser aging and decreased sensitivity of photodetectors. However, these methods rely heavily on precise physical parameters and complex mathematical derivations, making it difficult to comprehensively account for the diverse and complex factors present in real-world operating environments, thereby limiting prediction accuracy.

With the rise of machine learning and deep learning technologies, data-driven prediction methods have emerged. These methods mainly rely on the analysis of the Digital Diagnostic Monitoring (DDM) of the optical module, which is a technology used in optical modules to enable users to monitor the real-time parameters of the optical module. For example, Long Short-Term Memory (LSTM) networks have been employed to capture the temporal characteristics of optical module operating data, and supervised detection schemes have been proposed as alternatives to traditional time-series forecasting methods [3]–[5]. Nevertheless, many of these studies are based on simulated data or small-scale experimental datasets, which differ significantly from the large-scale, complex, and heterogeneous real-world data found in intelligent computing centers. As a result, these models often exhibit limited gener-

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alization capability in practical applications.

This paper is rooted in the real-world operations of intelligent computing centers, utilizing extensive and varied optical module data for training. It covers the entire lifecycle of different module models across various service scenarios. The paper introduces a new approach by combining advanced deep learning algorithms like LSTM and Transformer models to create a high-precision optical module lifetime prediction model. This aims to fill gaps in data authenticity and feature completeness found in current research, offering practical theoretical insights and technical solutions for the intelligent operation and maintenance of optical modules in computing center networks.

The main contributions are summarized as follows.

- We advance research by collecting extensive, real-life data from intelligent computing centers on various optical module models. This approach, using authentic lifecycle data, improves model generalization in complex situations, offering a solid foundation for accurate life predictions.
- We enhance model performance by integrating LSTM for short-term dynamics and Transformer for long-distance dependencies, effectively extracting multi-level features from operational data. This approach significantly boosts prediction accuracy and reliability in intelligent computing networks.

## II. RELATED WORK

In recent years, numerous researchers have proposed various machine learning and deep learning methods based on DDM to predict failures in optical modules.

In terms of machine learning-based methods, Wang *et al.* [6] proposed a machine learning method based on double exponential smoothing (DES) and SVM. They used DES to calculate the future DDM of the module, and then used SVM to determine the failure state of the module. Smalakys *et al.* [7] used Bayesian statistics to predict the remaining lifespan of optical components. Due to the intrinsic constraints of machine learning, these approaches demonstrate only moderate efficiency and accuracy.

With regard to deep learning-oriented approaches, Pei *et al.* [8] proposed a cross-timestep fault prediction method for optical modules in Internet data centers, which combines DDM data with port traffic information, uses an attention mechanism to address long-distance dependency issues, and mitigates data imbalance through an improved loss function.

However, among the aforementioned methods, none have taken into account the dual nature of the DDM data of optical modules. Specifically, it exhibits both long-distance dependencies and short-term dynamic variation characteristics.

## III. PRELIMINARIES

In this section, we first collect the data of optical modules in the cluster network, including seven types of data such as transmit optical power, receive optical power, bias current,

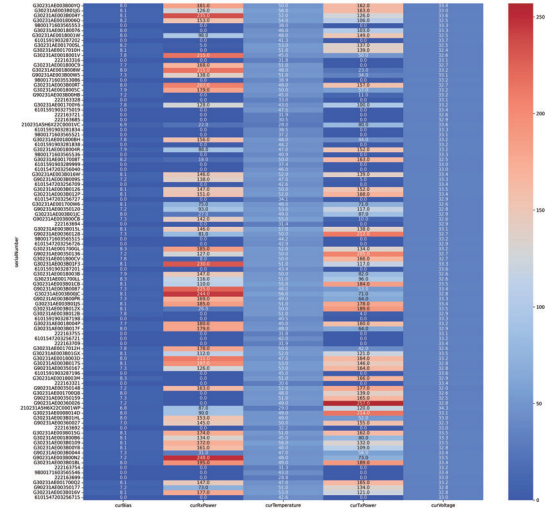


Figure 1. Visualization chart of optical module data.

temperature, bias voltage, number of packet errors, and operating status. Based on these 7 types of data, we conduct a visual analysis, as shown in Figure 1. Finally, we conclude that transmit optical power, receive optical power, bias current, and temperature are the main factors affecting the operating status of optical modules.

## IV. DESIGN OF PREDICTION MODEL

In this section, we designed a Transformer-LSTM-based model to predict the remaining useful life of optical modules and assess their health status accurately. This model leverages the strengths of both Transformer and LSTM. LSTM effectively captures long-term dependencies in time series data, whereas Transformer uses self-attention mechanisms to grasp global sequence information.

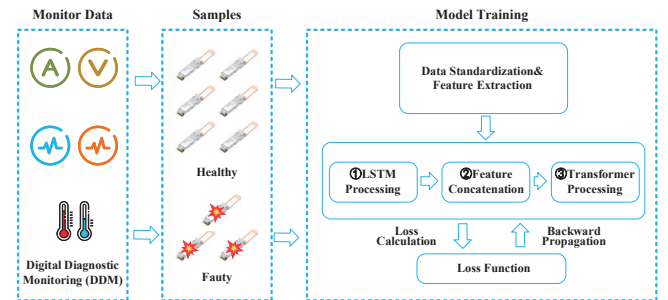


Figure 2. Workflow of Transformer-LSTM based fault prediction model.

As shown in Figure 2, it is the structural diagram of the Transformer-LSTM based fault prediction model for optical modules. Since the input data for optical module degradation are multiple time series data, including the DDM data of optical modules (operating voltage, operating temperature, received power, transmitted power, and bias current) and port throughput, while the output data is a one-dimensional result of optical module degradation value, this paper optimizes and

designs the traditional LSTM prediction model to obtain a multi-input single-output regression prediction model. Multi-input means that the model can receive multiple time series data of different types or from different sources, and single-output refers to the prediction of one-dimensional variables (such as optical module degradation values).

In the following section, the architecture of the proposed model is detailed. The model consists of six layers organized in a sequential manner.

- **Input Layer:** This layer receives multiple time-series inputs with varying dimensions. The input data is structured as [Batch, Seq, Dim], where Batch indicates the batch size, Seq stands for the sequence length, and Dim represents the feature dimension.
- **LSTM Layers:** Each input sequence is processed independently by an LSTM layer with 64 hidden units, which helps extract temporal dependencies within the sequence. The output shape of each LSTM layer is [Batch, Seq, 64].
- **Feature Concatenation Layer:** The outputs from the LSTM layers are concatenated along the feature dimension, creating a combined feature vector of size  $64 \times 3 = 192$ . The output shape of this layer is [Batch, Seq, 192].
- **Transformer Encoder Layer:** This layer uses self-attention mechanisms to capture global dependencies across the sequence, processing the concatenated features. Configured with a hidden dimension of 192 and 4 attention heads, the Transformer encoder retains the output shape of [Batch, Seq, 192].
- **Fully Connected Layer:** This layer transforms the Transformer encoder's output into a single regression value, reducing the feature dimension from 192 to 1. The output shape is [Batch, 1].
- **Output Layer:** This layer generates the final regression prediction, indicating the estimated degradation level of the optical module.

This architecture combines the strengths of LSTM and Transformer components. The LSTM layers capture local temporal patterns in individual input streams, while the Transformer encoder identifies global interdependencies across the combined feature space. The resulting hybrid model efficiently processes multi-modal time-series data, making it ideal for predicting the degradation of optical modules.

## V. PERFORMANCE EVALUATION

In this section, we evaluate the performance of the proposed Transformer-LSTM-based model for predicting the remaining useful life of optical modules. Given the involvement of time series data, Mean Absolute Error (MAE) and Mean Squared Error (MSE) are selected as evaluation metrics. In the subsequent content, we first introduce the dataset used for model training, followed by an explanation of the adopted evaluation metrics, and finally present the evaluation results of the model.

### A. Datasets

The datasets used in this paper are derived from the monitoring data of the intelligent computing center in Zhejiang Lab over the past year. The entire dataset consists of 500,000 negative samples and 5,00 positive samples.

### B. Experimental Settings

The program runs in a laptop with a 2.20GHz\*4 Intel Core i7-3632QM processor, 8GB memory, and Ubuntu 16.04 operating system. In the model training phase, datasets are split into a training set, a validation set, and a test set, with a ratio of 70%, 15%, and 15% respectively. During the training process, the training data is input into the model in batches, the loss is calculated, and backpropagation is performed to update the model parameters. Multiple rounds of training are conducted, and the model accuracy is evaluated using the validation set. The model is considered converged when its performance on the validation set no longer improves for several consecutive epochs.

### C. Metrics

**Prediction Time:** Prediction time refers to the time taken for a prediction algorithm to execute, and it is used to measure the prediction efficiency of the algorithm.

**Accuracy:** Accuracy is used to measure the degree of matching between the model's predicted results and the actual labels, with the calculation formula as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

**MSE:** Mean Squared Error (MSE) is used to calculate the average of the squares of errors between predicted values and actual values, with the calculation formula as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

where  $n$  denotes the number of samples,  $y_i$  represents the true value of the  $i$ -th sample, and  $\hat{y}_i$  refers predicted value.

**MAE:** Mean Absolute Error (MAE) is also used to calculate the average of the absolute errors between predicted values and actual values, with the calculation formula as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

where the meanings of the letters in the formula are similar to those in MSE.

### D. Experimental Results

Figure 3 shows the relationship between prediction time and the number of optical modules. It can be seen from the figure that as the number of optical modules increases, the prediction time also grows slowly. Compared with the traditional manual troubleshooting for a single optical module failure, which takes hours, when the number of optical modules reaches 100,000,

the overall prediction time of the proposed mechanism is about 6.5 seconds, which indicates that the prediction algorithm has high prediction efficiency.

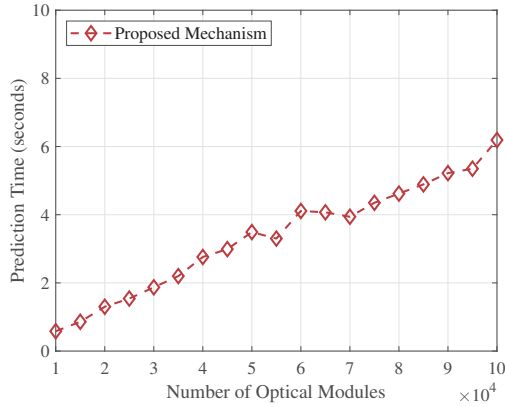


Figure 3. Prediction time of optical modules.

The comparison of accuracy, MSE, and MAE among different models, as shown in Table. I. Experimental results indicate that the designed model outperforms models such as SVM and CNN in terms of indicators like accuracy, MSE, and MAE. This is because although the DDM data of optical modules are time-series data, they have high complexity and contain many nonlinear relationships. The LSTM-Transformer model has strong advantages in capturing long-range dependencies and complex nonlinear relationships.

|                  | Accuracy | MSE      | MAE    |
|------------------|----------|----------|--------|
| LSTM-Transformer | 95.28    | 0.000963 | 0.0187 |
| CNN              | 73.56    | 0.003521 | 0.0412 |
| SVM              | 61.89    | 0.004785 | 0.0519 |

## VI. CONCLUSION

In this paper, we propose a data-driven remaining useful life prediction scheme for optical modules applicable to intelligent computing networks. Specifically, by combining the long-sequence dependency modeling capability of Transformer with

the advantage of LSTM in capturing temporal features, we design a Transformer-LSTM-based prediction model for wireless timeout faults in train control systems, which significantly enhances the early warning capability before faults occur. Finally, the model performance is verified through metrics such as accuracy, MAE, and MSE, and the results demonstrate that the designed model has favorable performance.

However, datasets comes from a single cluster with a limited number of models of optical modules, thus the generalization performance of the model also has certain limitations. In the future, more other types of optical modules need to be incorporated to further enhance the generalization performance of the model.

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